

January 30, 2015

Industrial waste test research completion report

The test and research based on the industrial waste test and research plan dated December 27, 2021 has been completed.

We would like to report on the results of industrial waste and treatment as follows.

Remember

Type of industrial waste Planned	Planned Quantity	Quantity obtained	Quantity used	Remaining Quantity	Treatment of residue (carbides)
Waste Plastic (single material)	5,000 kg	380 kg	250 kg	130 kg	The entire 35.1 kg of charred material was appropriately disposed of as cinders
waste plastic (plastic containers, etc..)	5,000 kg	150 kg	150 kg	0 kg	The entire 16.8 kg of charred material was appropriately disposed of as cinders
Shredder dust (mixture of waste plastics, glassscraps, ceramic scraps and metal scraps)	2,000 kg	171 kg	70 kg	101 kg	The entire 14.4 kg of charred material was appropriately disposed of as cinders
Total	12,000 kg	701 kg	470 kg	231 kg	Total quantity of residue (carbide) 66.3 kg

*1: Regarding the remaining 130 kg of waste plastics (single material) that were not used in the test and research due to the test and research schedule.

was promptly returned to the supplier, Tatsuzaki Co., Ltd., after the test research was completed.

The above waste plastics (single material), which are originally valuable materials, were provided free of charge for testing and research purposes.

*2: Waste plastics (plastic containers, etc.) are waste generated within the [REDACTED] Factory, and only the required quantity is used.

There was no remaining amount because I obtained it.

*3: Shredder dust was judged to be unsuitable for oil processing due to the extremely poor properties of the resulting oil.

Dardust returned the product to its supplier, Yamatake Tsuchiura Sales Office, as soon as the testing and research was completed.

Industrial waste test research report

September 2022

, Ltd.

Joint conductor of industrial waste test and research

One World Co., Ltd.: Pyrolysis equipment manufacturer/operation management

Land Brain Co., Ltd.: Various measurements and analyzis

Table of contents

Table of Contents

1.	Purpose of the test and research.....	1
2.	Contents of test research	1
3.	Location of the test and research	2
4.	Test research process	5
5.	About used industrial waste sourced	9
6.	Waste plastic pyrolysis oil test	13
7.	Removal of produced oil	23
8.	Items confirmed in test research	33
9.	Regarding product disposal	37
10.	Analysis results of produced oil and oil residue (carbide).....	39

1. Purpose of the test and research

We conducted a demonstration test using the superheated steam decomposition recycled resource recovery processing equipment "URBAN RIG" manufactured by One World Co., Ltd., and evaluated the quality and quality of products (produced oil, charcoal, etc.) manufactured by this equipment. The purpose of this project is to evaluate and confirm the environmental load generated by this device, and to contribute to solving the problem of plastic waste recycling, which will become increasingly important in the future.

2. Contents of test research

Matters to be confirmed in test and research.

Requirements as an oil conversion facility Based on the Act on the Enforcement of "URBAN RIG", the facility "URBAN RIG" satisfies the structure and method as a pyrolysis facility. Therefore, through this test and research, we will evaluate whether or not this equipment was properly processed using the structure and method as a "pyrolysis equipment" shown in attachment 1.

The quality of the product will be confirmed in order to consider the intended use and sales destination of the products (produced oil, carbide, etc..) produced by this equipment.

Clarify the quality and value of produced oils and charcoal products, taking into account their calorie content, physical composition, and harmful substance content.

Environmental load confirm whether the emissions (exhaust gas, wastewater, noise, vibration, etc.) generated by this equipment meet the regulatory standards in Japan.

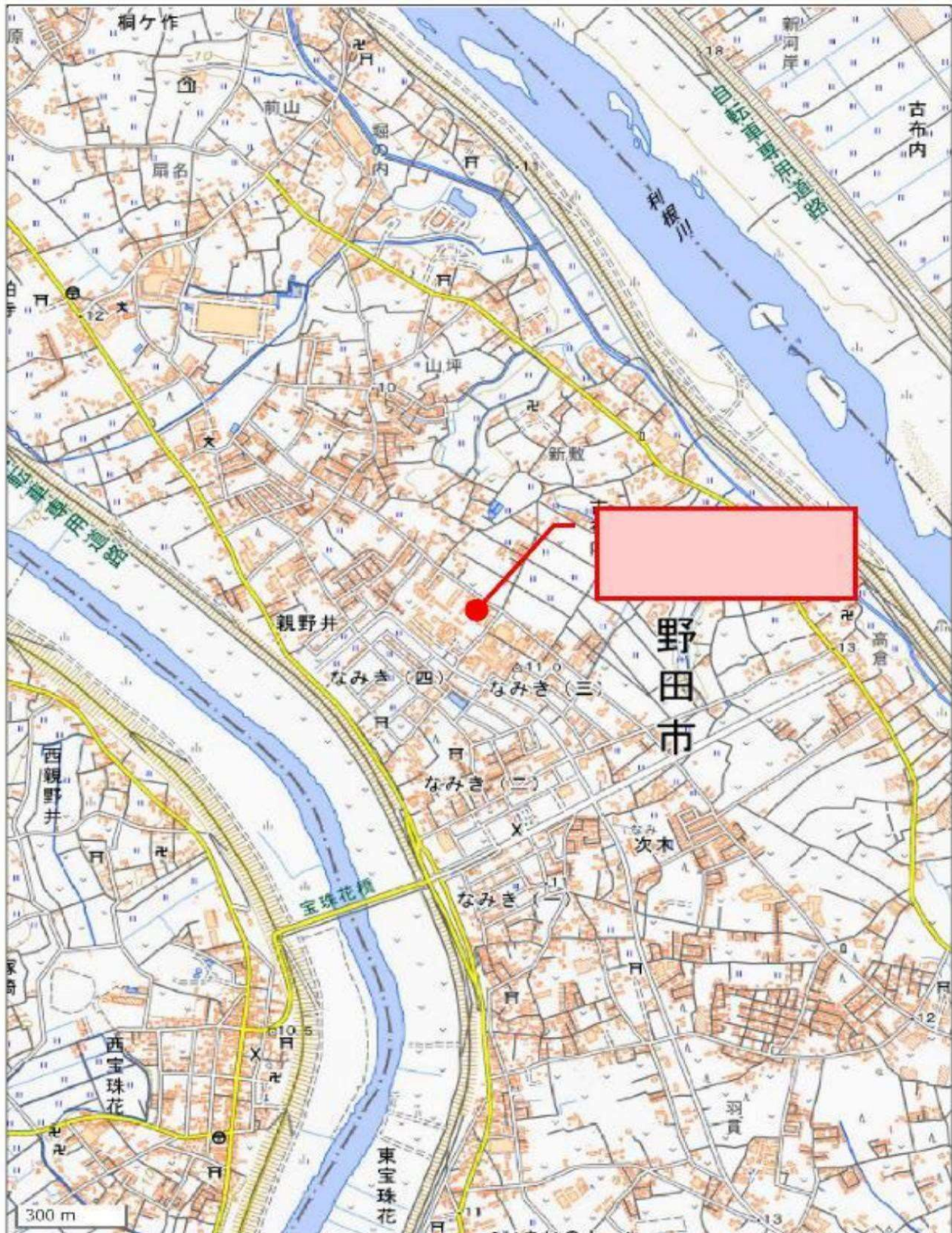
The measurement and analysis results are compared with emission standards such as the Air Pollution Control Act and the Water Pollution Control Act, standard values such as the Noise Control Act and Vibration Control Act, the regulatory standards of the Offensive Odor Control Act, and the regulatory standards of the Noda City Environmental Conservation Ordinance and clarify whether there are any issues.

Material balance Material balance, such as the weight and chemical composition, contained components, and retained heat of the input waste, including the weight and chemical composition, contained components, and retained heat of the generated carbides, hydrocarbon oil, off-gas, combustion exhaust gas.

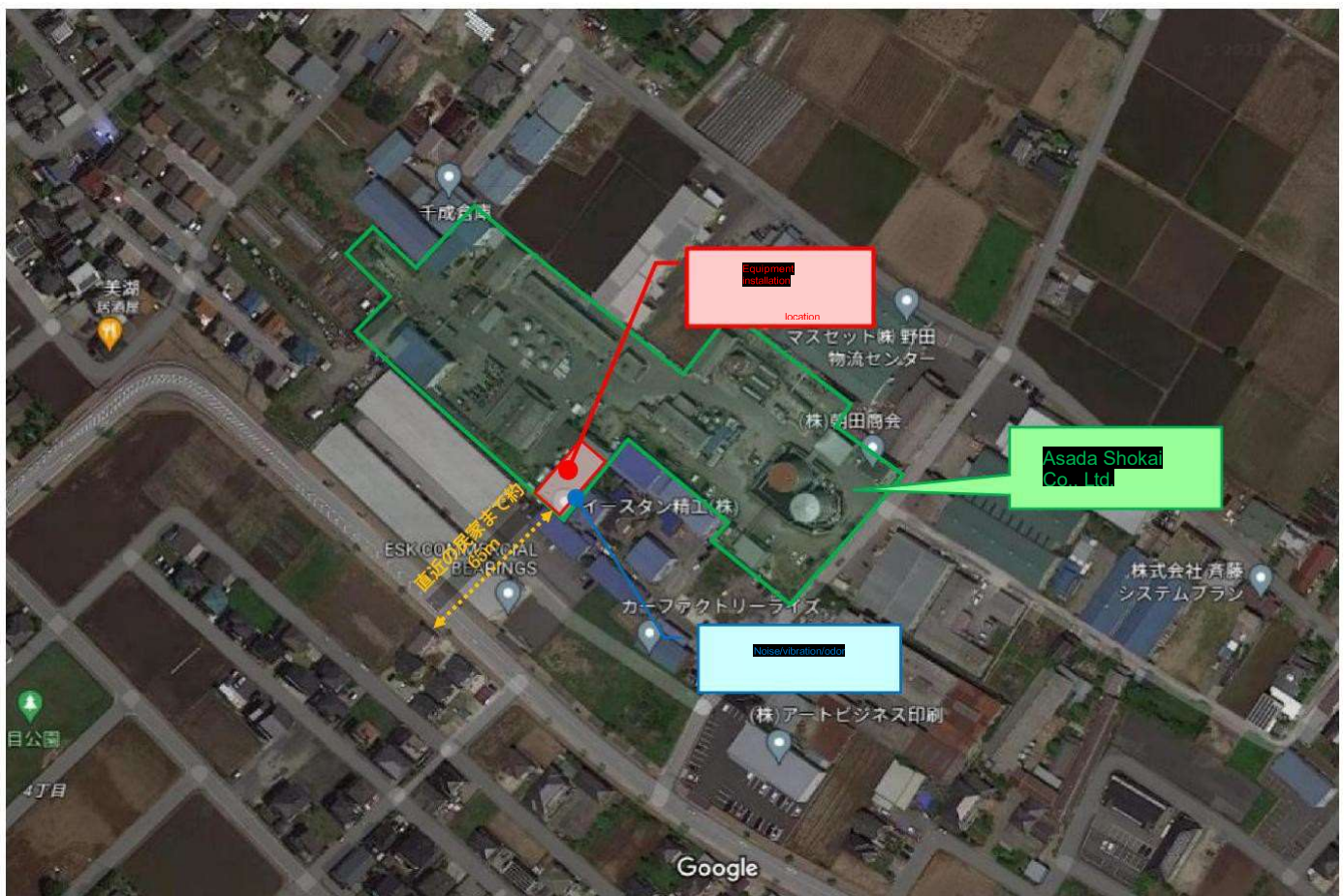
3. Location of the test and research

Noda City, Chiba Prefecture

(1) Surrounding floor plan



(2) Surrounding aerial photograph



(3) Campus layout plan

Confidential

4. Test research process

From March 2 2020 to September 30 2014

For details, see material 1 industrial waste test research schedule on page 7.

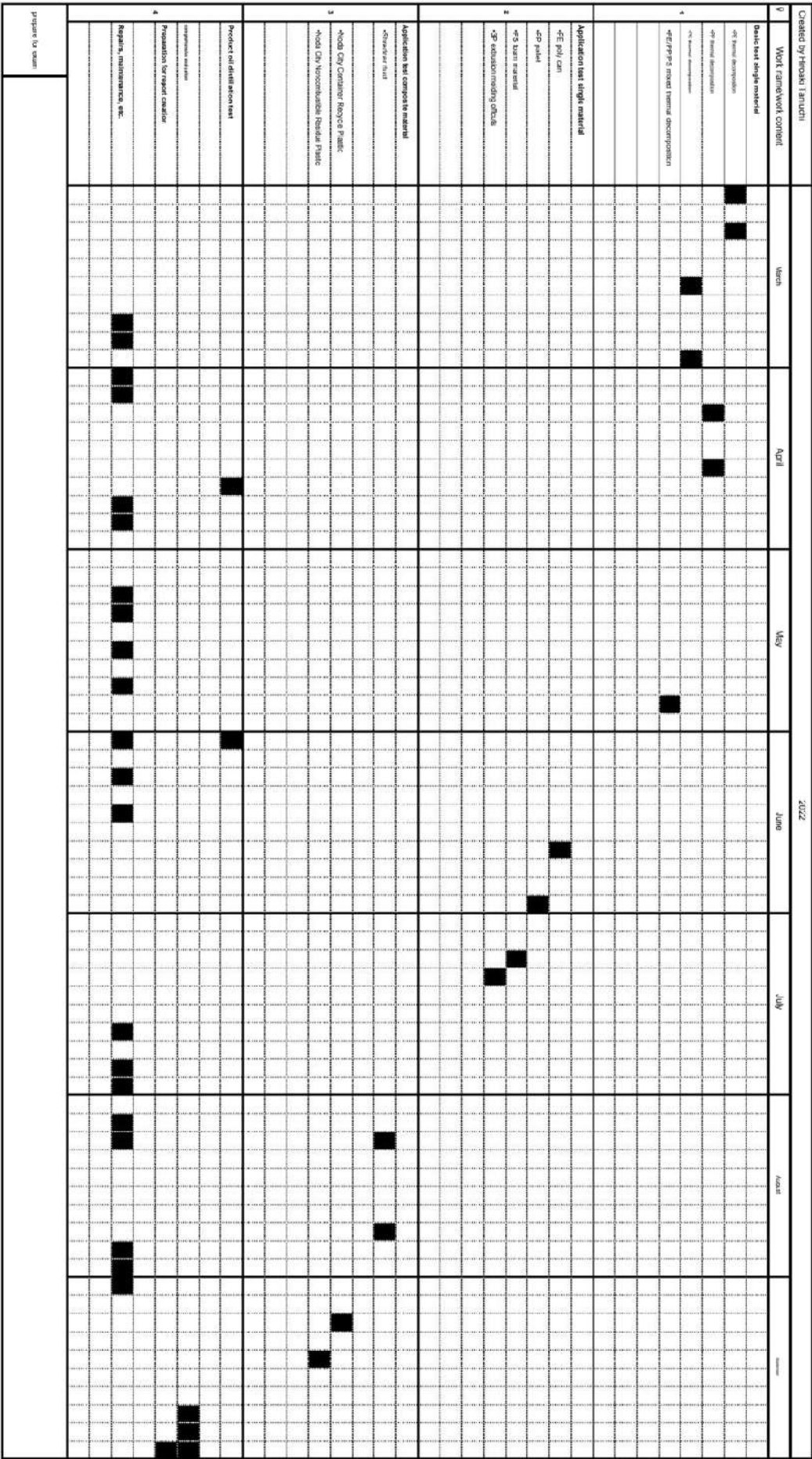
Data 1: Engineering table

Construction equipment	Industrial waste test and research
Special Equipment vehicles	Asada Shokai Mota Factory
Business partner	Pro Wood Co. Ltd./ and Iron Co. Ltd.

Industrial waste test research process chart

Execution period	Start	March 2, 2022
	End	September 30, 2022

Test conductor	Asada Shokai Co., Ltd.
Production date	March 2, 2022
Person in charge	Hirotaki Taniguchi
	5



5. About used industrial waste sourced

(1) Single material for basic testing

Source: Tatsuzaki Co., Ltd.

Location: 306-11 Kirigasaku, Noda City, Chiba Prefecture



PE pellets



PE molding scraps



PP pellets



PP molded scraps



PS pellets



PS molded scraps

(2) Single material for waste test 1

Source: Asada Shokai Co., Ltd.

Address: 3-22-1 Namiki, Noda City, Chiba Prefecture



PE poly can



PE poly can



PP plastic pallet



PP plastic pallet



PS Styrofoam



PS Styrofoam

(3) Composite material for waste test 2

Source: Yamatake Tsuchiura Sales Office

Location: 5303-4 Kamo, Kasumigaura City, Ibaraki Prefecture



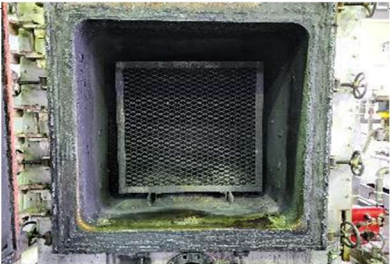
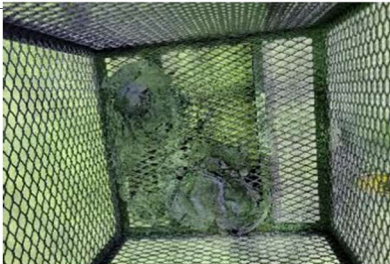
shredder dust



shredder dust

6. Waste plastic pyrolysis oil test

2022/3/2	Loaded weight	20 Kg	Produced oil weight	14.8 Kg	Oil conversion rate	74%
	Residual weight	0.3 Kg	Volume of oil produced	18 liters	Off gas rate	25%
	Metal / foreign residue w	0 kg	Produced oil specific gravity	0.82		
	Oil residue weight	0.3 Kg	Off gas amount	4.9 kg	Drainage amount	62 liters
PE Pellet basic test						
						
Inputs						
end of process status						

2022/3/9	Loaded weight	40 Kg	Produced oil weight	30.7 Kg	Oil conversion rate	77%
	Residual weight	0.5 kg	Volume of oil produced	37 liters	Off gas rate	22%
	Metal / foreign residue w	0 Kg	Produced oil specific gravity	0.83		
	Oil residue weight	0.5 Kg	Off gas amount	8.8 kg	Drainage amount	64 liters
PE molded scraps basic test						
						
Inputs			Preparation for launch			
end of process status			Residue condition			

2022/3/17	Loaded weight	20 kg	Produced oil weight	12.2 kg	Oil conversion rate	61%
	Reisdual weight	3.8 kg	Volume of oil produced	14 liters	Off gas rate	20%
	Metal / foreign residue w	0 kg	Produced oil specific gravity	0.87		
PS pellet basic test	Oil residue weight	3.8 kg	Off gas amount	4.0 kg	Drainage amount	46 liters
Inputs			Preparation for launch			
end of process status			Residue condition			

2022/3/31	Loaded weight	40 kg	Produced oil weight	22.0 kg	Oil conversion rate	55%
	Reisdual weight	15.3 kg	Volume of oil produced	25 liters	Off gas rate	7%
	Metal / foreign residue w	0 kg	Produced oil specific gravity	0.88		
PS molded scrap basic test	Oil residue weight	15.3 kg	Off gas amount	2.7 kg	Drainage amount	55 liters
Inputs			Preparation for launch			
end of process status			Residue condition			

2022/4/8	Loaded weight	20 kg	Produced oil weight	16.2 kg	Oil conversion rate	81 %
	Reisdual weight	0.5 kg	Volume of oil produced	19 liters	Off gas rate	17 %
PP Pellet basic test	Metal / foreign residue w	0 kg	Produced oil specific gravity	0.85		
	Oil residue weight	0.5 kg	Off gas amount	3.3 kg	Drainage amount	78 liters
Inputs			Preparation for launch			
end of process status			Residue condition			

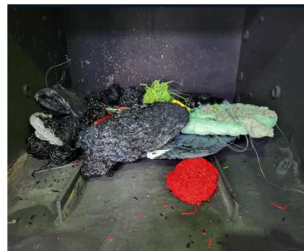
2022/4/16	Loaded weight	50 kg	Produced oil weight	39.8 kg	Oil conversion rate	80 %
	Reisdual weight	0.2 kg	Volume of oil produced	48 liters	Off gas rate	20 %
PP molded scrap basic test	Metal / foreign residue w	0 kg	Produced oil specific gravity	0.83		
	Oil residue weight	0.2 kg	Off gas amount	10.0 kg	Drainage amount	90 liters
Inputs			Preparation for launch			
end of process status			Residue condition			

2022/5/26	Loaded weight	60 kg	Produced oil weight	43.1 kg	Oil conversion rate	72 %
	Reisdual weight	5.2 kg	Volume of oil produced	51 liters	Off gas rate	19 %
	Metal / foreign residue weight	0 kg	Produced oil specific gravity	0.845		
	Oil residue weight	5.2 kg	Off gas amount	11.7 kg	Drainage amount	73 liters
Inputs			Preparation for launch			
end of process status			Residue condition			

2022/6/21	Loaded weight	50 kg	Produced oil weight	40.7 kg	Oil conversion rate	81 %
	Reisdual weight	0.8 kg	Volume of oil produced	47 liters	Off gas rate	17 %
	Metal / foreign residue weight	0 kg	Produced oil specific gravity	0.867		
	Oil residue weight	0.8 kg	Off gas amount	8.4 kg	Drainage amount	73 liters
Inputs			Preparation for launch			
end of process status			Residue condition			

2022/6/28	Loaded weight	50 kg	Produced oil weight	36.5 kg	Oil conversion rate	73 %
	Residual weight	6.1 kg	Volume of oil produced	45 liters	Off gas rate	15 %
PP Pallet Waste Test Single Material	Metal / foreign residue w	0 kg	Produced oil specific gravity	0.812		
	Oil residue weight	6.1 kg	Off gas amount	7.4 kg	Drainage amount	73 liters
Inputs			Preparation for launch			
end of process status			Residue condition			

2022/7/7	Loaded weight	50 kg	Produced oil weight	36.3 kg	Oil conversion rate	73 %
	Residual weight	9.9 kg	Volume of oil produced	43 liters	Off gas rate	8 %
PS Foam material waste test single	Metal / foreign residue w	0 kg	Produced oil specific gravity	0.845		
	Oil residue weight	9.9 kg	Off gas amount	3.8 kg	Drainage amount	88 liters
Inputs			Preparation for launch			
end of process status			Residue condition			

2022/7/11		Loaded weight	60.0 kg	Produced oil weight	49.4 kg	Oil conversion rate	82 %
		Reisdual weight	9.3 kg	Volume of oil produced	60 liters	Off gas rate	2 %
3P molded offcuts basic test		Metal / foreign residue weight	0 kg	Produced oil specific gravity	0.823		
		Oil residue weight	9.3 kg	Off gas amount	1.3 kg	Drainage amount	85 liters
Inputs				Preparation for launch			
	end of process status				Residue condition		

2022/8/9		Loaded weight	50 kg	Produced oil weight	0 kg	Oil conversion rate	-
		Reisdual weight	9.8 kg	Volume of oil produced	0 liters	Off gas rate	-
Shredder Dust Waste Test Composite Mat'l	Metal / foreign residue weight			Produced oil specific gravity	0		
	Oil residue weight		9.8 kg	Off gas amount	-	Drainage amount	73 liters
Inputs				Preparation for launch			
end of process status				Residue condition			

2022/8/22		Loaded weight	20 kg	Produced oil weight	0 kg	Oil conversion rate	-
		Reisdual weight	4.6 kg	Volume of oil produced	0 liters	Off gas rate	-
		Metal / foreign residue weight	-	Produced oil specific gravity	0		
		Oil residue weight	4.6 kg	Off gas amount	-	Drainage amount	71 liters
Inputs				Preparation for launch			
end of process status				Residue condition			

2022/9/9		Loaded weight	50 kg	Produced oil weight	16.1 kg	Oil conversion rate	41 %
		Reisdual weight	24.4 kg	Volume of oil produced	18 liters	Off gas rate	24 %
		Metal / foreign residue weight	10.5 kg	Produced oil specific gravity	0.874		
		Oil residue weight	13.9 kg	Off gas amount	9.5 kg	Drainage amount	52 liters
Inputs			Preparation for launch				
end of process status			Residue condition				

2022/9/15		Loaded weight	50 kg	Produced oil weight	22.5 kg	Oil conversion rate	50 %
		Reisdual weight	16.7 kg	Volume of oil produced	26 liters	Off gas rate	24 %
		Metal / foreign residue weight	5.4 kg	Produced oil specific gravity	0.864		
		Oil residue weight	11.3 kg	Off gas amount	10.8 kg	Drainage amount	52 liters
Inputs			Preparation for launch				
end of process status			Residue condition				

Oil conversion rate: The ratio of the weight of the produced oil to the input weight minus the weight of inorganic materials such as metals and glass that do not contribute to oil conversion.

Off-gas amount: Value obtained by subtracting the removed weight and produced oil weight from the input weight.

Off-gas rate: The ratio of the amount of off-gas to the input weight minus the weight of metals and foreign substances that do not contribute to oil conversion.

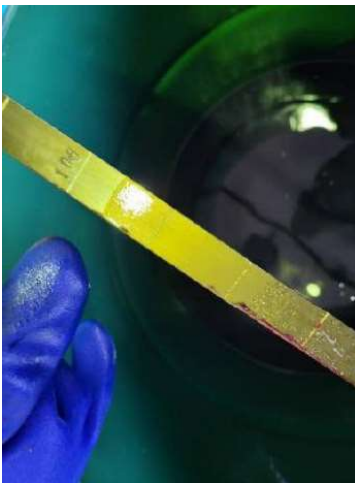
7. Removal of produced oil

(1) Measurement of produced oil



The oil produced in the oil-water separation layer

The gas generated in the pyrolysis furnace passes through a catalytic reaction tank, is condensed in a condenser, passes through a drain pot, and is led to an oil-water separation layer.



Normally, the produced oil is taken out after draining only the water from the drain of the oil-water separation layer, but in this test and research, both the oil and water were transferred from the oil-water separation layer to a drum using a pump, and after static separation in the drum, a measuring rod was removed. was used to measure the amount of oil and water.

First, measure the total quantity of oil and water.



Then, measure the amount of water using Water Feeling Paste, which turns red when it comes in contact with water.

Oil is measured by subtracting the measured amount of water from the total measured amount

(2) Distillation of produced oil



Generated oil (3P (PE, PP, PS))

Since this test equipment does not have a process for measuring the produced oil, the produced oil obtained is a mixture of the first petroleum, the second petroleum, the third petroleum, etc.

First petroleum products are difficult to handle, and there is no established market for recycled products. On the other hand, second class petroleum and above can be expected to be reused as recycled products. Here, the obtained product oil was fractionated using a distillation apparatus into first class petroleum and second class petroleum.



distillation equipment



Produced oil set in distillation equipment



distilled first petroleum

The following are oils produced from virgin materials (PE, PP, and PS alone and 3P, which is a mixture of PE, PP, and PS) that are fractionated and are classified into the first class petroleum, the second class petroleum, and above.



Distillation test of PE-generated oil

Because PE oil contains a large amount of wax, its fluidity decreases when the outside temperature drops.



Distillation test of PP produced oil PP

Backwater oil has high fluidity, and even after fractionating the first petroleum, the produced oil of second or higher petroleum class maintains its fluidity.



Distillation test of PS produced oil

PS contains a lot of carbon in its molecules, so it becomes a dark brown produced oil. After separating the first petroleum, the resulting oil of the second petroleum or higher has low fluidity.



Distillation test of 3P produced oil

The 3P produced oil is in a state that includes the characteristics of PE, PP, and PS. This is the state closest to the oil produced in plastic containers and packaging.

プラスチック熱分解生成油蒸留・分析結果

原料	P.P.		P.E.		P.S.		3P(PE,PPS)	
	油温度(°C)	蒸気温度(°C)	油温度(°C)	蒸気温度(°C)	油温度(°C)	蒸気温度(°C)	油温度(°C)	蒸気温度(°C)
初留	100	30	110	45	150	57	98	50
5mℓ			138	57	154	112	120	57
10mℓ	115	60	145	70	155	117	125	57
20mℓ			165	105	157	123	130	70
30mℓ	139	83	175	122	161	126	135	118
40mℓ				135	163	127	153	134
50mℓ	163	105		135	166	130	159	137
60mℓ			235	150	169	133	167	145
70mℓ	170	138			174	135	176	154
80mℓ						136	188	163
90mℓ						138	195(85mℓ)	168(85mℓ)
100mℓ						143		
処理量	200mℓ		200mℓ		200mℓ		200mℓ	
蒸留量	第一石油類	第二石油類以下	第一石油類	第二石油類以下	第一石油類	第二石油類以下	第一石油類	第二石油類以下
	56mℓ	114mℓ	61mℓ	139mℓ	90mℓ	110mℓ	96mℓ	104mℓ
引火点	18.0°C	45.0°C	18.0°C	55.0°C	16.0°C	53.0°C	16.0°C	52.5°C
比重	0.760mℓ/ g	0.820mℓ/ g	0.756mℓ/ g	0.860mℓ/ g	0.870mℓ/ g	0.900mℓ/ g	0.788mℓ/ g	0.834mℓ/ g
硫黄分	30.4ppm	101.5ppm	19.2ppm	52.4ppm	58.1ppm	147.2ppm	50.5ppm	77.3ppm

(3) Properties of produced oil, Impurities, etc.



Produced oil immediately after removal contains water and other impurities.

The precipitate can be seen at the bottom of the beaker in the photo above.



Produced oil after heat treatment

By subjecting the extracted produced oil to superheating treatment, it becomes easy to separate impurities, etc., and it is possible to obtain highly purified produced oil.

(4) Evaluation of produced oil



Produced oil obtained from plastic containers and packaging

A dark brown product oil was obtained compared to the product oil from virgin material. Perhaps because it contains a lot of polyethylene as a raw material, it contains a lot of wax, and its fluidity decreases as the temperature drops.



Produced oil obtained from incombustible residue

Compared to produced oil for containers and packaging plastic, it contains more impurities, but it has higher fluidity than plastic produced oil for containers and packaging, probably because the proportion of polyethylene is lower.

The oil obtained from the thermal decomposition of plastics is a mixture of so-called primary petroleum, which has a light specific gravity and a low flash point, and secondary and tertiary petroleum, which have relatively high specific gravity.

In the industry that manufactures recycled oil from waste oil, the uses of primary petroleum products are limited, and in order to commercialize the produced oil as recycled oil, these primary petroleum products must be fractionated using distillation equipment.

8. Items confirmed in test research

(1) Requirements for an oil conversion facility

Thermal decomposition (oil conversion) tests for containers and packaging plastics and incombustible residues are conducted in such a way that the weight of the hydrocarbon oil produced is 40% of the weight of the treated waste.

In addition, the weight of gas generated during processing that is not recovered as hydrocarbon oil was 25% or less of the weight of treated waste, and the requirements for an oil processing facility were met.

(2) Product quality

Produced oil

As mentioned earlier, produced oil is produced as a mixture of first class petroleum and hydrocarbon oil of second class petroleum or higher. These produced oils, especially the first class petroleum oils, have many restrictions under the Fire Service Act, so by fractionating the first class petroleum oil using distillation equipment, etc., it may be possible to effectively use it as recycled fuel such as recycled oil. In addition, since the distribution price of the separated first petroleum is extremely low using existing methods, it is necessary to separately develop new uses for it as a raw material for petroleum products.

Oil residue (carbide)

The properties of oil residue (carbide) vary greatly depending on the content of the object to be treated and the temperature and time of the oil treatment. It is necessary to control the temperature and time appropriately for each object to be treated to suppress the generation of carbide, as well as understand the properties of the carbide and consider how to use it effectively.

(3) Environmental load

Exhaust gas

The concentration of soot and dust in the exhaust gas is 0.003g/Nm^3 at the furnace temperature of $550\text{ }^{\circ}\text{C}$, sulfur oxides are less than 0.005 g/Nm^3 , nitrogen oxides are 43ppm, and dioxins are 0.067ng-TEQ/ Nm^3 .

The measurement result was m3, both of which were below the regulatory values. *See page 49 for details.

Drainage

Since this test device does not have a wastewater treatment process, the analysis results of the raw water before wastewater treatment exceeded the water quality standard values. Based on this, it will be reflected in the wastewater treatment equipment of practical equipment. The sampling dates were September 9th, Noda City's repura treatment day, and September 15th, Noda City's non-combustible residue treatment day, both of which were selected as days when it was thought that the greatest load would be placed on wastewater. *See page 45 for details

Noise and vibration

Noise and vibration levels are both below the regulatory standards at the site boundary, so there are no problems. *See page 53 for details.

Odors

There are no problems with the 22 odour items at the site boundaries, all of which are below regulatory standard values. *See page 53 for details.

(4) Material Balance/Energy Balance

Material balance and energy_balance were calculated based on the case where the 3P pellets on May 26th and the 3P molding offcuts on July 11th were treated to oil.

The energy balance ratio was 77.4% when processing 3P pellets and 80% when using 3P molded scraps. This was a fair value since it was a process for processing objects with relatively little foreign matter or dirt, **but further improvements in energy efficiency are required for practical use.**

For reference, the energy balance when processing recycled plastic containers and packaging and non-flammable residue in Noda City was 40.7% and 52.5%.

Improving oil energy efficiency this will lead to improvements not only in the conversion rate but also in business viability. Regarding the transpiration water of water (drainage) in material balance, the calculated value was used. *See page 35 for details.

(5) Main issues

The pyrolysis equipment used for this test and research was a test equipment, so it was not able to fully demonstrate its processing capacity. Additionally, it took time to repair and maintain the superheated steam system, burners, etc. due to malfunctions, and it was not possible to process the amount of industrial waste that was originally planned. In the future, we plan to present the issues identified in this demonstration experiment to the manufacturer (One World) and proceed with product development of the improved processing equipment ``Superheated Steam Decomposition Recycled Resource Recovery Processing Equipment." .

[Main issues]

- Improving processing capacity
- Improving energy balance
- Dealing with mixed plastics containing PET resin
- Installing fractionation equipment

Material 2 :Material Balance/Energy Balance

input into processing			
	Product name	Input	Energy input
1	Electricity	65 kwh	55,900 kcal
2	Light Oil	66 liters	599,808 kcal
3	Off gas	11.7 kg	15,489 kcal
4	Total energy input		671,197 kcal
5	Water	85 liters	

Raw Material		
	Product Name	Total input amount
1	May 26th	60 kg
	3P (PE, PP,PS) Pellets	

Products			
	Product Name	Generation	Energy Generation
1	generated Oil	43.1 kg	473.238 kcal
2	Oil Residu	5.2 kg	30,576 kcal
	metal / foreign matter	0.0 kg	
	Total residue	5.2 kg	
3	Off gas	11.7 kg	15,489 kcal
4	Total products	60 kg	519,303 kcal
5	drain water	73 liters	
	Water vapor	12 liters	
	Total water	85 liters	

Material Balance Ratio		100%
Energy Balance Ratio		77.4%

input into processing			
	Product name	Input	Energy input
1	Electricity	65 kwh	55,900 kcal
2	Light Oil	76 liters	690,688 kcal
3	Off gas	1.3 kg	1,721 kcal
4	Total energy input		748,309 kcal
5	Water	95 liters	

Raw Material		
	Product Name	Total input amount
1	July 11th	60 kg
	3P molded offcuts	

Products			
	Product Name	Generation	Energy Generation
1	generated Oil	49.4 kg	542,412 kcal
2	Oil Residu	9.3 kg	54,684 kcal
	metal / foreign matter	0.0 kg	
	Total residue	9.3 kg	
3	Off gas	1.3 kg	1,721 kcal
4	Total products	60 kg	598,817 kcal
5	drain water	85 liters	
	Water vapor	10 liters	
	Total water	95 liters	

Material Balance Ratio		100%
Energy Balance Ratio		80.0%

Reference material : City material balance of recycled plastic containers and packaging as well as non combustible residue.

input into processing			
	Product name	Input	Energy input
1	Electricity	65 kwh	50,740 kcal
2	Light Oil	61 liters	554,368 kcal
3	Off gas	9,5 kg	12,576 kcal
4	Total energy input		617,684 kcal
5	Water	100 liters	

Raw Material		
	Product Name	Total input amount
1	Sept. 9th	50 kg
	mixed waste	

Products			
	Product Name	Generation	Energy Generation
1	generated Oil	16.1 kg	176,778 kcal
2	Oil Residu	10.5 kg	61,740 kcal
	metal / foreign matter	13.9 kg	
	Total residue	24.4 kg	
3	Off gas	9.5 kg	12,576 kcal
4	Total products	50 kg	251,094 kcal
5	drain water	73 liters	
	Water vapor	27 liters	
	Total water	100 liters	

input into processing			
	Product name	Input	Energy input
1	Electricity	65 kwh	55.900 kcal
2	Light Oil	61 liters	554,368 kcal
3	Off gas	10.8 kg	14,297 kcal
4	Total energy input		624,565 kcal
5	Water	100 liters	

Raw Material		
	Product Name	Total input amount
1	Sept, 15th	50 kg
	mixed waste	

Products			
	Product Name	Generation	Energy Generation
1	generated Oil	22.5 kg	542,412 kcal
2	Oil Residu	11.3 kg	54,684 kcal
	metal / foreign matter	5.4 kg	
	Total residue	16.7 kg	
3	Off gas	10.8 kg	14,297 kcal
4	Total products	50 kg	327,791 kcal
5	drain water	85 liters	
	Water vapor	15 liters	
	Total water	100 liters	

Material Balance Ratio		100%
Energy Balance Ratio		52.5%

9. Regarding product disposal

(1) Disposal method and destination of the remaining amount of waste subjected to oil processing

Single resin scraps (resin pellets, molded product scraps, etc.)

Supplier: Tatsuzaki Co., Ltd. (306-11 Kirigasaku, Noda City, Chiba Prefecture)

	PP		PP		PS		3P	Total
	Pellet	Molded Scrap	Pellet	Molded Scrap	Pellet	Molded Scrap	Molded Scrap	
Planned Amount								5000 kg
Carried amount	40 kg	60 kg	40 kg	60 kg	40 kg	60 kg	80 kg	380 kg
Processed amount	20 kg	40 kg	20 kg	40 kg	20 kg	50 kg	60 kg	250 kg
Remaining amount	20 kg	20 kg	20 kg	20 kg	20 kg	10 kg	20 kg	130 kg

Due to the test and research schedule, pellets and molded product scraps that were not used in the test and research will be discarded at the end of the test and research.

Immediately after the completion of the project, it was returned to Tatsuzaki Co., Ltd.

The above pellets and molded product scraps, which are inherently valuable, were provided free of charge for testing and research purposes.

Waste plastics (waste plastic containers, etc.)

	PE poly can	PP pallet	PS foam	Total
Measure				
Amount input	50 kg	50 kg	50 kg	150 kg
Amount processed	50 kg	50 kg	50 kg	150 kg
Remaining	0 kg	0 kg	0 kg	0 kg

The above waste is generated within the premises, so please only use the necessary quantity. There is no remaining amount as it was procured and used for testing and research.

Industrial waste (shredder dust)

Regarding the shredder dust mentioned above, since there is no example of processing it with this device in the past, we will first conduct a pre-test. We decided to implement the following. We procured 171kg of shredder dust and processed 50kg and 20kg of it into oil, but given the properties of the resulting oil, it was determined that the shredder dust was in very poor condition and was not suitable for oil processing, and the remaining 101kg of shredder dust It was returned to Yamatake Tsuchiura Sales Office, the supplier

10. Analysis results of produced oil and oil residue (carbide)

test report of plastic produced oil in containers and packaging

2022年10月31日 11時26分

一般社団法人 潤滑油協会

NO. 5238 P. 4

試験成績表

TEST RESULTS

試験報告番号

TEST REPORT NO. 20221020-003-03

株式会社朝田商会 御中

一般社団法人 潤滑油協会

JAPAN LUBRICATING OIL SOCIETY

〒273-0015 船橋市日の出 2-4-2

TEL : 047-433-5181 FAX : 047-431-9572

日付

DATE 2022年10月31日



下記のとおりご報告申し上げます。

試験項目	試料名	客リプラ生成油		
密度 (15℃) g/cm ³		0.8642 ^{※1}		
反応		酸性		
引火点 (P.M.) °C		40 未満 ^{※2}		
動粘度 (50℃) mm ² /s		1.706		
流動点 °C		10.0		
残留炭素分 質量%		0.34		
水分 (電量滴定法) 質量%		0.13		
灰分 質量%		0.002		
硫黄分 質量%		0.05 未満		
窒素分 質量%		0.24		
塩素分 質量%		0.137		
総発熱量 (計算) J/g		46000		
摘 要 ※1 密度は、JIS K 2249-2 浮ひよう法による。 ※2 測定条件 : 21.5℃、湿度 61%、102.3KPa				

2022年10月31日 11時26分

一般社団法人 潤滑油協会

NO. 5238 P. 5

試験成績表

TEST RESULTS

試験報告番号

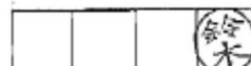
TEST REPORT NO. 20221020-003-04

株式会社朝田商会 御中

一般社団法人 潤滑油協会
 JAPAN LUBRICATING OIL SOCIETY
 〒273-0015 船橋市日の出 2-1-1
 TEL : 047-433-5181 FAX : 047-431-9529

日付

DATE 2022年10月31日



下記のとおりご報告申し上げます。

試験項目	試料名	残渣プラ生成油		
密度 (15℃) g/cm ³		0.8762 ^{※1}		
反応		酸性		
引火点 (P.M.) °C		40 未満 ^{※2}		
動粘度 (50℃) mm ² /s		1.759		
流動点 °C		15.0		
残留炭素分 質量%		0.44		
水分 (電量滴定法) 質量%		0.20		
灰分 質量%		0.014		
硫黄分 質量%		0.06		
窒素分 質量%		0.43		
塩素分 質量%		0.526		
総発熱量 (計算) J/g		45000		
摘要 ※1 密度は、JIS K 2249-2 浮ひよう法による。 ※2 測定条件：21.5℃、湿度 61%、102.3KPa				

分析結果報告書

No.F2210007 - 1

ランドブレイン株式会社 様

発行 2022 年 09 月 22 日

受 付 2022 年 09 月 09 日
 試料採取 2022 年 09 月 09 日 自社
 試験実施 2022 年 09 月 09 日 ~ 2022 年 09 月 21 日
 提供試料 生成炭化物 (含有量試験)
 件 名 廃プラ油化装置実証調査

株式会社オオスミ
 登録番号 神奈川 濃度第18号
 横浜市瀬谷区五貫目町 20番地 17
 〒246-0008 電話045 (923) 1056 (代)
 FAX 045 (923) 1058
 環境計量士 小谷 智樹

検査承認	試験実施

上記提供試料の分析の結果は下記のとおりであることを報告します。

分 析 の 対 象	生成炭化物	-----	-----	試 験 方 法
カドミウム	31 mg/kg			底質調査方法 (平成24年 環境省) II. 5. 1. 1
六価クロム	0.25 mg/kg 未満			底質調査方法 (平成24年 環境省) II. 5. 12. 3
シアン化合物	0.5 mg/kg 未満			底質調査方法 (平成24年 環境省) II. 4. 11. 1
総水銀	0.06 mg/kg			底質調査方法 (平成24年 環境省) II. 5. 14. 1. 1
鉛	140 mg/kg			底質調査方法 (平成24年 環境省) II. 5. 2. 1
砒素	0.95 mg/kg			底質調査方法 (平成24年 環境省) II. 5. 9. 2
ふっ素化合物	130 mg/kg			底質調査方法 (平成24年 環境省) II. 4. 12. 2
ほう素	36 mg/kg			底質調査方法 (平成24年 環境省) II. 5. 12. 3
セレン	0.5 mg/kg 未満			底質調査方法 (平成24年 環境省) II. 5. 10. 1
- 以 下 余 白 -				
採取時刻 水 天 気 温 候 考 値 考	有姿ベース			

分析結果報告書

No.F2210007-1

ランドブレイン株式会社 様

発行 2022 年 09 月 22 日

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 横浜市瀬谷区五貫目町 30番地-17
 〒246-0008 電話045(920)1050-111
 FAX 045(920)1050-111
 環境計量士 小谷 智樹



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分 析 の 対 象	生成炭化物	-----	-----	試 験 方 法
カドミウム	31 mg/kg			底質調査方法(平成24年環境省) Ⅱ.5.1.1
六価クロム	0.25 mg/kg 未満			底質調査方法(平成24年環境省) Ⅱ.5.12.3
シアン化合物	0.5 mg/kg 未満			底質調査方法(平成24年環境省) Ⅱ.4.11.1
総水銀	0.06 mg/kg			底質調査方法(平成24年環境省) Ⅱ.5.14.1.1
鉛	140 mg/kg			底質調査方法(平成24年環境省) Ⅱ.5.2.1
砒素	0.95 mg/kg			底質調査方法(平成24年環境省) Ⅱ.5.9.2
ふっ素化合物	130 mg/kg			底質調査方法(平成24年環境省) Ⅱ.4.12.2
ほう素	36 mg/kg			底質調査方法(平成24年環境省) Ⅱ.5.13.2
セレン	0.5 mg/kg 未満			底質調査方法(平成24年環境省) Ⅱ.5.19.1
- 以 下 全 白 -				
採取時刻	—			
水 温	—			
天 候	—			
気 温	—			
備 考	有姿ベース			

